MILITARY SPECIFICATION SHEET

SWITCH, WAVEGUIDE, ELECTROMECHANICAL, FAILSAFE
FREQUENCY RANGE 8.5 - 10.3 GHz

This specification sheet is approved for use by the Engineering and Technical Support Division, AFLC/LOIE, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the switch described herein shall consist of this document and the latest issue of Military Specification MIL-S-55041

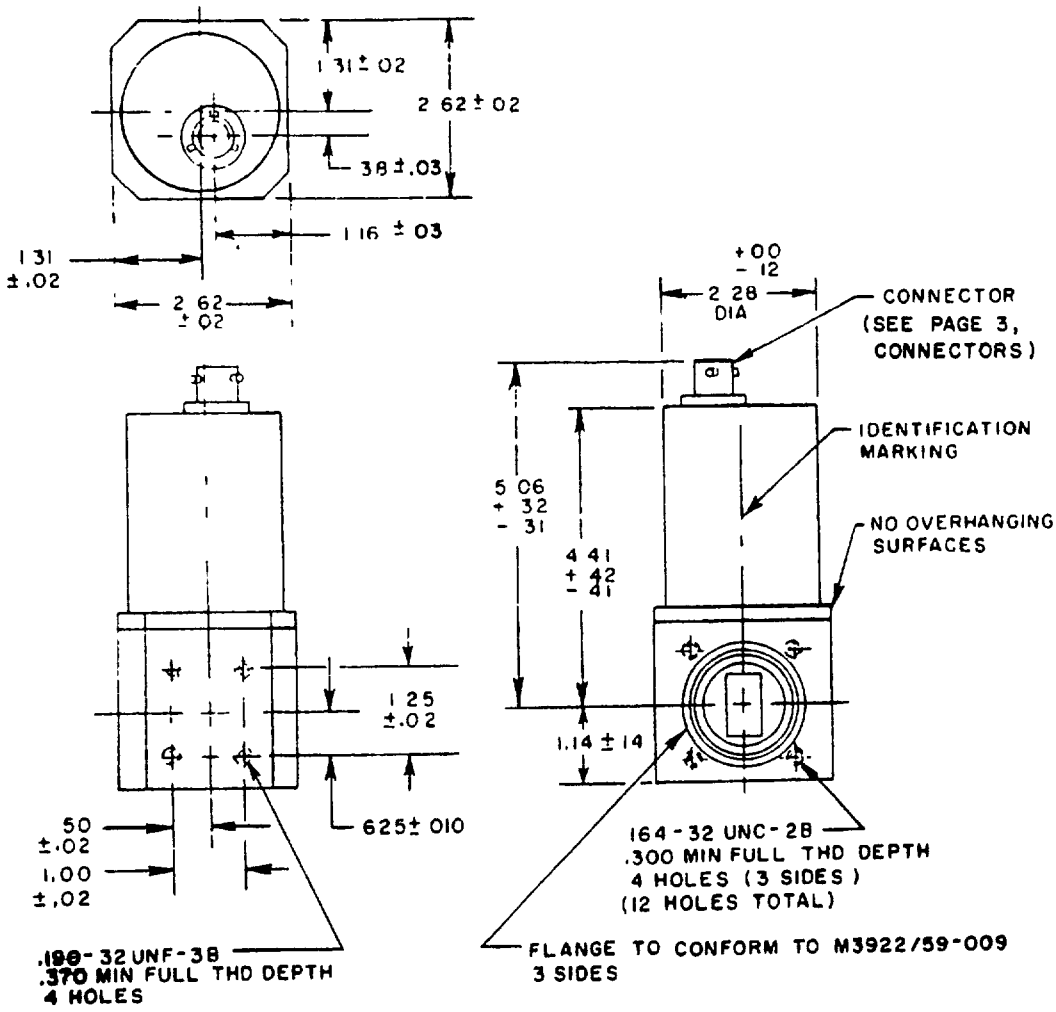
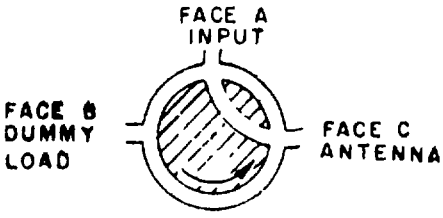


FIGURE 1 Switch configuration and schematic, part number M55041/5-001.

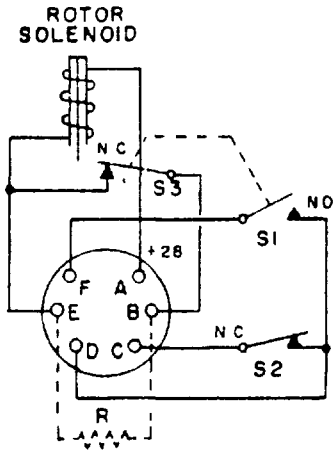
FSC 5985

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:18Z
Check the source to verify that this is the current version before use.



DIRECTION OF ROTOR ROTATION IS DETERMINED BY LOOKING DOWN ON THE SWITCH FROM ITS TOP

INCHES	MM
.010	.25
.02	.5
.03	.8
.12	3.1
.14	36
.164	4.17
.190	4 83
.300	7.62
31	7.9
.32	8.1
.370	9 40
.38	9.7
41	10 4
.42	10.7
.50	12 7
625	15.88
1 00	25.4
1 14	29.0
1 16	29.5
1 25	31.8
1 31	33 3
2. 28	57 1
2 62	66.6
4 41	112.0
5 06	128.5



EXTERNAL SOLENOID HOLDING 'R' RESISTOR - 20 OHM 20 WATTS (NOT SUPPLIED)

SHOWN IN DEENERGIZED POSITION

SCHEMATIC DIAGRAM

NOTE: Energized: The circuit C-D (S2) shall open within 100 of the rotor travel before F-D (S1) is closed (within .100 of final rotor travel).
Deenergized: F-D ust open prior to C-D being closed (within .100 of final rotor travel).

FIGURE 1. Switch configuration and schematic, part number M55041/5-001-Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:18Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Design and construction:
Dimensions and configuration: See figure 1.
Equipment class: The waveguide switch shall be in accordance with MIL-E-5400, class 2.
Solenoid: See figures 1 and 2.
Weight: 40 ounces (1.134 kg), maximum.
Temperature range: -55 to +100°C.
Connectors: The connector shall be MS3723/08HICC6N (flange mount type) or MS3723/12HICC6N (jam nut type), per MIL-C-83723.
Exterior surface: Black per MIL-F-7179.

Pull-in voltage: 24 Vdc.

Drop-out voltage: 16 Vdc.

Operating current: 4 amperes, maximum at 27 Vdc.

Holding current: 1 ampere, maximum, at 27 Vdc.

Isolation: 80 dB, minimum.

VSWR: 1.10:1, Maximum

Insertion loss: 0.20 dB, maximum.

Switching time: Switch shall be mounted in the position shown in figure 3. The switching time shall be taken and recorded (maximum 1.0 second). The switch shall be rotated in 45° increments, and switching time shall be recorded at each position through 360°. The switching time shall not vary more than 10 percent from the time recorded at 0° (the original position) nor exceed 1.0 second.

RF power handling capability: Peak, 400 kW; average, 400 W.

Cycle life: 50,000 cycles at the rate of 10 cycles per minute.

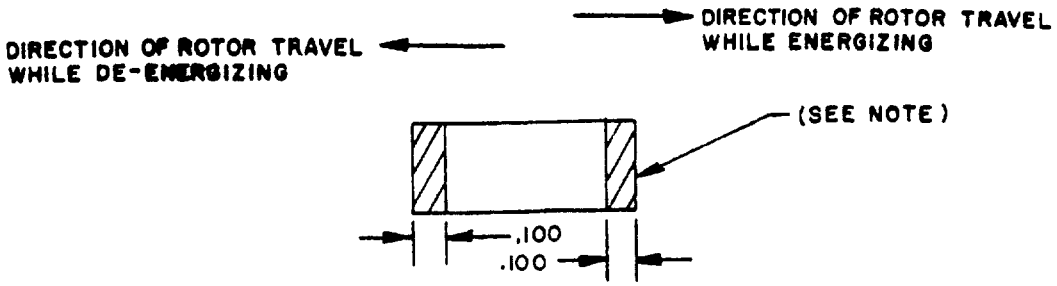
Thermal shock: Test condition A; step 3 shall be 100°C.

Temperature altitude: 1/ Switches shall be tested in accordance with MIL-STD-810, method 504.1, procedure I for equipment category 6. The following details apply:
a. Initial measurements: Not applicable.
b. Equipment operation: RF need not be applied. Only the solenoid and interlock functions are energized.
c. Within 10 minutes after completion of this test, isolation, VSWR, insertion loss, holding current, pull-in voltage, drop-out voltage, and switching time shall be measured.

Pressure equilibrium: When compressed air is applied at any waveguide port, the pressure differential shall not exceed 0.5 pound. The following details apply:
a. Seal one active (open) waveguide port and attach a compressed air source of 15 lbf/in^2 (gauge) to the other active port.
b. Adjust air flow rate to 150 cubic inches per minute.
c. Measure the difference between the active ports and the closed port.

1/ This test shall be substituted for altitude and cold test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:18Z
Check the source to verify that this is the current version before use.



NOTE. Shaded area is visible portion of rotor when viewing face A
The rotor is shown in two positions

FIGURE 2 Interlock circuit and rotor travel

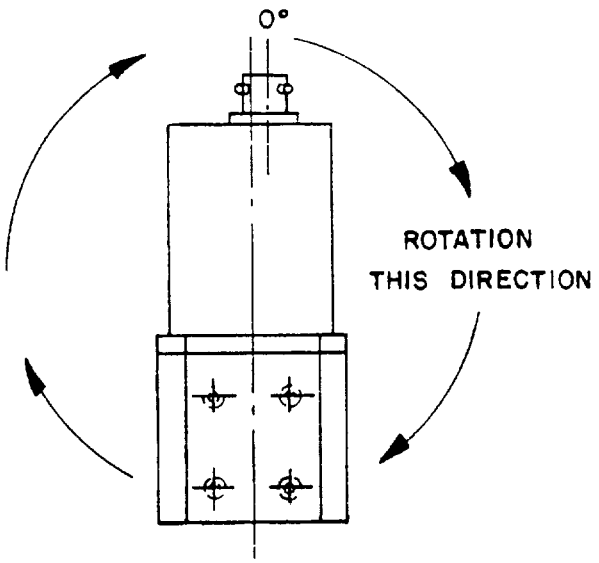


FIGURE 3 Switching time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:18Z
Check the source to verify that this is the current version before use.

Vibration, high frequency: 2/ The switch shall be tested in accordance with method 204 of MIL-STD-202, test condition C. The switch shall be firmly secured to the vibration platform without the use of shock isolators. During vibration the switch shall operate at the rate of not less than one cycle per minute. The switching circuit and interlock switches shall be monitored to determine if any malfunction occurs. Following the test, isolation, VSWR, insertion loss, holding current, pull-in voltage, drop-out voltage, switching time, and rotor-motion delay shall be measured and shall meet the specified requirements.

Part number: M55041/5-001.

Substitutability: This switch is substitutable for USAF P/N 7334755, NSN 5985-00-821-5636.

2/ This test shall be substituted for the vibration test.

Custodian:
Air Force - 85

Preparing activity:
Air Force - 85

Review activity:
Air Force - 99

(Project 5985-F464)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:18Z
Check the source to verify that this is the current version before use.

FOLD

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEPARTMENT OF THE AIR FORCE
DoD-318

AFLC/LOIE/Standardization
Wright-Patterson AFB OH 45433

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:18Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-S-55041/5 (USAF) Switch, Waveguide, Electromechanical

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER FSC 5985 MATERIAL PROCURED UNDER A [ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

YES [ ] NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

C 1 6 0 F 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:18Z
Check the source to verify that this is the current version before use.